



Sharif University of Technology
Aerospace Engineering

Subject
Phase B2 of System Design Of 3U Precision
Agriculture CubeSat PRELIMINARY
DESIGN

Author
Reza Heidari
Seyed Reza Hoseinitabar

Instructor
Dr. Khabdakhsh

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Phase B2 Preliminary Design

Abstract

This report presents the **Phase B2 Preliminary Design** of the **AgriSat-3U** mission, a 3U CubeSat developed for precision agriculture through multispectral Earth observation and Normalized Difference Vegetation Index (NDVI) analysis. Building upon the system requirements established during the System Requirements Review (SRR), this phase transforms the approved functional architecture into a preliminary physical design through hardware selection, subsystem sizing, and system integration.

Formal engineering trade-off studies were conducted for the major spacecraft subsystems, including the Attitude Determination and Control System (ADCS), Electrical Power Subsystem (EPS), communications architecture, structural configuration, and payload processing hardware. Candidate solutions were evaluated using weighted decision matrices to identify the most suitable technologies while satisfying mission performance, reliability, mass, power, and implementation constraints.

Detailed analytical sizing was performed for the optical payload, electrical power subsystem, reaction wheels, communications subsystem, structural components, thermal control, onboard data storage, and spacecraft performance budgets. Mass, power, data, pointing, thermal, and communications link budgets were developed under worst-case operational scenarios defined by the mission Concept of Operations (ConOps), and engineering margins were evaluated against the allocated system requirements.

A complete preliminary system architecture is presented, including subsystem block diagrams, interface definitions, a hierarchical Product Tree and Bill of Materials (BOM), commercially available component (COTS) selection, and a Level-1 CAD configuration used to verify internal packaging, solar array placement, sensor field-of-view clearance, and subsystem integration. Project risks were systematically identified through a structured risk register with corresponding mitigation strategies, while compliance with applicable ECSS standards and Preliminary Design Review exit criteria was assessed.

The resulting preliminary design demonstrates that the selected spacecraft architecture satisfies the mission requirements with acceptable engineering margins and establishes a stable technical baseline for progression to the Critical Design Phase (Phase C), where detailed design, verification, manufacturing planning, and qualification activities will be completed.

Keywords: CubeSat, Preliminary Design Review, System Architecture, Trade-Off Analysis, Subsystem Sizing, ECSS, COTS Selection, Space Systems Engineering, Precision Agriculture, NDVI.

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1 Introduction

Following the successful completion of Phase B1 (Systems Engineering & Functional Architecture), this Phase B2 package develops the system design within the *Solution Space*. The focus is transitioned from "what" the spacecraft must do to "how" it will physically accomplish its mission objectives using flight-proven hardware.

The primary mission payload is a high-resolution multispectral pushbroom imager (RGB + Near-Infrared) designed to generate Normalized Difference Vegetation Index (NDVI) products for agricultural monitoring. This payload operates from a 650 km Sun-Synchronous Orbit (SSO) with a Local Time of Ascending Node (LTAN) of 10:30 AM.

This report establishes the physical baseline by:

1. Performing formal trade-off studies using Pugh decision matrices to select critical subsystem topologies.
2. Presenting mathematical sizing derivations for the Electrical Power Subsystem (EPS), Attitude Determination and Control Subsystem (ADCS), and Communications Subsystem under worst-case ConOps loads.
3. Detailing a hierarchical Product Tree and physical Bill of Materials (BOM) using industry-standard Commercial-Off-The-Shelf (COTS) CubeSat components.
4. Refining the system budgets (Mass, Power, RF Link, Data, and Pointing) with safety margins matching the European Cooperation for Space Standardization (ECSS) frameworks.
5. Documenting a Level 1 CAD configuration and layout check.

2 PROJECT BASELINE & SYSTEM REQUIREMENTS TRACEABILITY

The design baseline is driven by a hierarchy of System Requirements (SR) derived from the mission goals. Below is the complete, traceable Requirements Traceability Matrix (RTM).

Verification Methods: A (Analysis), I (Inspection), T (Test), D (Demonstration).

Table 2 1. System Requirements Traceability Matrix (RTM)

Req. ID	Requirement Description	Allocated Subsystem	Verification Method	Status	Design Margin / Parameter	Reference Section
SR-01	Total wet mass must not exceed 4.500 kg.	SMS	A, T	Compliant	5.75% mass margin (Dry: 4.21 kg, Wet: 4.241 kg)	7.1
SR-02	Structure must conform to standard 3U CubeSat specifications (100.0 x 100.0 x 340.5 mm).	SMS	I, T	Compliant	Fit clearance: 1.0 mm to deployer walls	8

SR-03	Mission design lifetime must be 3 to 5 years in LEO.	SYS / TCS	A	Compliant	Radiation analysis indicates <10 krad (Si) TID	9
SR-04	Ground sampling distance (GSD) must be ≤ 10 m in multispectral bands.	PAY	A, D	Compliant	GSD = 9.2 m at 650 km nadir	5.1
SR-05	Payload must cover a swath width of 20 km.	PAY	A	Compliant	Swath = 20.0 km at 650 km altitude	5.1
SR-06	Pointing accuracy must be 0.2° ($0.001^\circ/\text{s}$) during imaging.	ADCS	A, T	Compliant	RSS Pointing Error = 0.1044° (47.8% margin)	7.6
SR-07	Pointing drift rate must be $0.02^\circ/\text{s}$ during pushbroom imaging.	ADCS	A	Compliant	Drift rate = $0.005^\circ/\text{s}$ under closed loop	5.3

SR-08	Battery must support safe operations during a 35.38-minute eclipse.	EPS	A, T	Compliant	Battery capacity 46.8 Wh vs 8.55 Wh req. (447%)	5.2
SR-09	EPS must maintain a positive orbit-averaged energy balance.	EPS	A	Compliant	Peak generation : 28.7 W; OAPC: 5.17 W	7.2
SR-10	Onboard data storage must support at least 10 raw image acquisitions.	CDHS	I, D	Compliant	64 GB flash storage (supports 47 undownlinked passes)	7.3
SR-11	Payload data downlink rate must be 2 Mbps.	COM	T	Compliant	S-Band physical rate: 2.0 Mbps	5.4
SR-12	S-Band Link Margin must be 9 dB under	COM	A, T	Compliant	Link Margin = 9.06 dB at 10° elevation	7.4

	worst-case elevation.				(202% margin)	
SR-13	UHF transceiver must provide command uplink and telemetry beacon backup.	COM	T	Compliant	Uplink rate: 9.6 kbps FSK	6.1
SR-14	Primary structures must maintain structural integrity under a 10 g quasi-static load.	SMS	A, T	Compliant	Minimum safety factor = 1.85 (Al 6061-T6)	5.5
SR-15	Component temperatures must remain within protoflight limits.	TCS	A, T	Compliant	Battery kept within [0°C, 25°C] with active heaters	7.5

3 SYSTEM ARCHITECTURE & INTERFACE CONTROL (ICD)

The *AgriSat-3U* system architecture is built on a highly integrated 3U platform utilizing a centralized PC104 electrical backplane and dedicated high-speed lines for payload data handling.

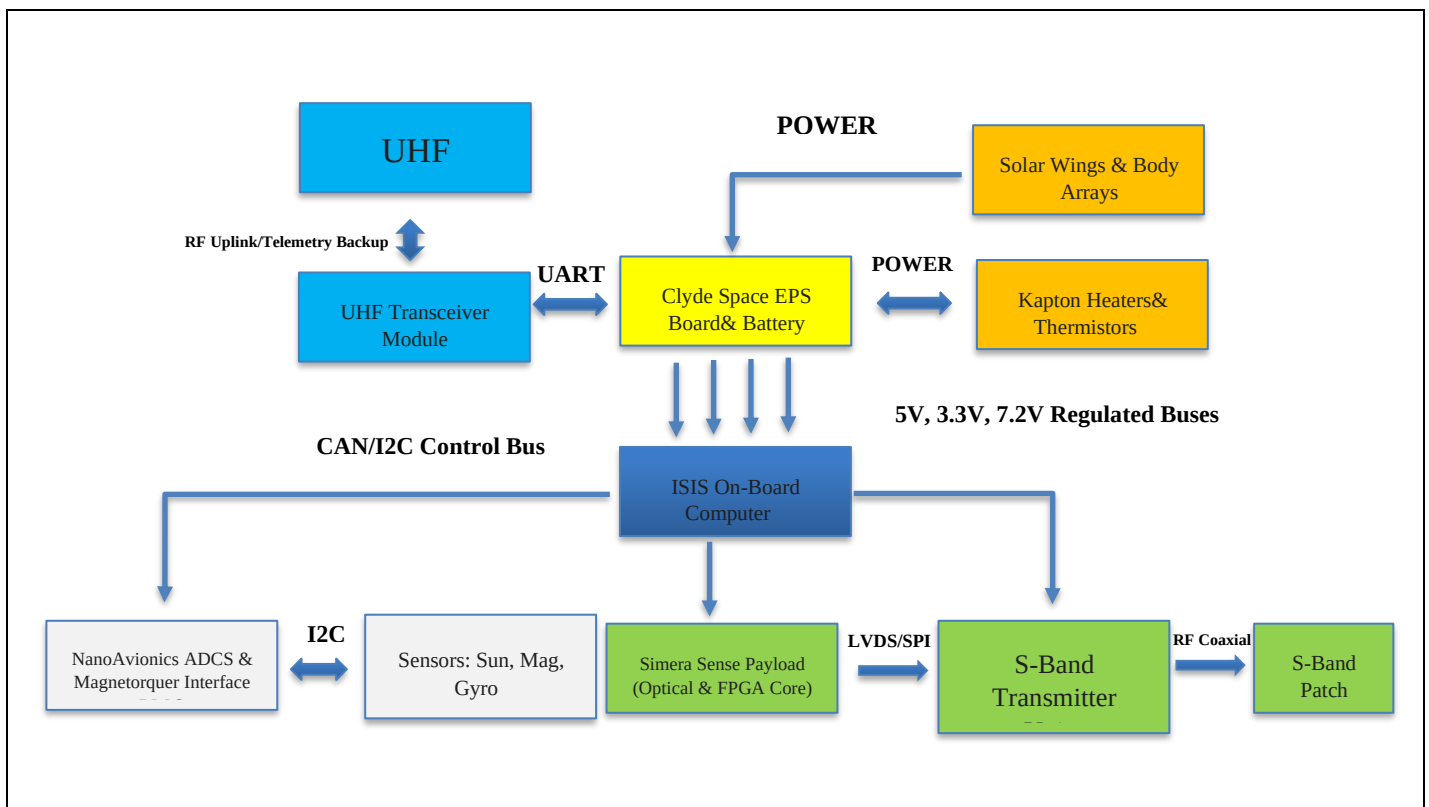


Figure 3 1. System Block Diagram and Inter-Subsystem Architectural Flows

3.1 Functional, Physical, and Logical Architectures

- Functional Architecture:** Maps the system capabilities. Command reception occurs via UHF, decoded by the CDHS, which schedules payload imaging sequences, commands the ADCS to execute a high-precision pitch-compensation slew, manages power allocation via the EPS, stores the high-speed imagery, and downlinks the processed results through the S-Band transmitter.

- **Physical Architecture:** Follows a standard stack configuration inside a Pumpkin 3U Monocoque Aluminum structure. Standard PC104 connectors route power and low-speed command lines. S-band high-speed data bypasses the shared PC104 bus via a dedicated, shielded LVDS/SPI cable directly from the Payload Processor to the S-band Transmitter to prevent bus-wide electromagnetic coupling and signal degradation.
- **Logical Architecture:** Operates on a prioritized state machine under a Real-Time Operating System (RTOS) on the ISIS OBC. State telemetry is collected at 1 Hz and recorded in a non-volatile memory partition.

3.2 Interface Control Document (ICD) Overview

To ensure error-free assembly, integration, and testing (AIT), physical interfaces are strictly constrained.

Table 3 1.System Inter-Subsystem Interface Mapping Matrix

Interface ID	Source Subsystem	Destination Subsystem	Interface Type	Physical Connection	Data Protocol / Voltage	Shielding / Isolation
IF-01	EPS	All Subsystems	Power	PC104 Bus	3.3V, 5V, 7.2V DC	Common ground
IF-02	CDHS	ADCS	Command /Data	PC104 Bus Pins	I2C / CAN	Ground reference isolators
IF-03	CDHS	EPS	Telemetry	PC104 Bus Pins	I2C (100 kHz)	None
IF-04	PAY	CDHS	Control	SAMTEC Connector	SPI / UART	Direct board-to-board
IF-05	PAY	COM (S-Band)	High-speed Data	Custom LVDS Harness	RS-422 / LVDS	Braided copper shield

IF-06	COM (UHF)	CDHS	Commands	PC104 Bus Pins	UART (115.2 kbps)	Optocouplers present
IF-07	COM (S-Band)	S-Band Patch Ant.	RF Payload	SMA Coaxial	S-Band RF, 50 Ω	Low-loss semi-rigid coax
IF-08	TCS	EPS / Heaters	Power / Analog	Custom Micro-D	Analog Thermistor / 7.2V	Kapton thermal isolation

4 Engineering Trade Study

To guarantee that design decisions are grounded in objective engineering metrics rather than qualitative estimation, multi-criteria trade studies with sensitivity analyses were performed.

4.1 ADCS Actuator Topology Trade-off

The pointing stability requirement ($0.02^\circ/\text{s}$ drift) is a hard driver. We trade three candidates:

1. **Option A:** Pure Magnetic Control (3x Magnetorquers)
2. **Option B:** Pure Momentum Control (3-Axis Reaction Wheels)
3. **Option C:** Hybrid Active Control (3-Axis Reaction Wheels + 3x Magnetorquers)

Table 4 1.ADCS Actuator Decision Matrix

Evaluation Criteria	Weight	Option A (Magnetic)	Option B (Pure Wheels)	Option C (Hybrid)
Pointing Accuracy (deg)	0.30	1 (Score: 0.30)	4 (Score: 1.20)	5 (Score: 1.50)
Pointing Stability (deg/s)	0.25	1 (Score: 0.25)	4 (Score: 1.00)	5 (Score: 1.25)
Momentum Management	0.20	5 (Score: 1.00)	1 (Score: 0.20)	5 (Score: 1.00)
System Mass & Vol. Impact	0.10	5 (Score: 0.50)	2 (Score: 0.20)	2 (Score: 0.20)
Power Consumption	0.10	4 (Score: 0.40)	2 (Score: 0.20)	2 (Score: 0.20)
Technical Heritage	0.05	5 (Score: 0.25)	3 (Score: 0.15)	4 (Score: 0.20)

Weighted Sum	1.00	2.70	2.95	4.35
Rank		3	2	1

Sensitivity Analysis: Even if mass and power constraints are weighted 50% higher, Option C's pointing advantages make it the only mathematically viable choice to satisfy SR-06 and SR-07. Pure wheels saturate within 1.08 orbits under aerodiag disturbance torque if un-dumped. Hybrid Active Control (Option C) is selected.

4.2 EPS Solar Array Configuration Trade-off

1. **Option A:** Fixed Body-Mounted Panels
2. **Option B:** Dual Deployable Tracking Arrays
3. **Option C:** Hybrid Configuration (Body-mounted + 2 Fixed Deployable Wings)

Table 4 2.EPS Solar Array Configuration Decision Matrix

Evaluation Criteria	Weight	Option A (Fixed Body)	Option B (Deployable Track)	Option C (Hybrid Fixed-Dep)
Peak Power Generation	0.35	2 (Score: 0.70)	5 (Score: 1.75)	4 (Score: 1.40)
Mechanism Reliability	0.25	5 (Score: 1.25)	1 (Score: 0.25)	4 (Score: 1.00)
Jitter & ADCS Coupling	0.20	5 (Score: 1.00)	1 (Score: 0.20)	4 (Score: 0.80)
Structural Complexity	0.10	5 (Score: 0.50)	1 (Score: 0.10)	3 (Score: 0.30)
Thermal Stability	0.10	4 (Score: 0.40)	2 (Score: 0.20)	3 (Score: 0.30)
Weighted Sum	1.00	3.85	2.50	3.80
Rank		1	3	2 (Selected)

Critical Analysis: Though Option A scores slightly higher due to mechanical simplicity, it cannot physically generate the 21.54 W required by the worst-case power sizing. Option C is the selected engineering baseline because it is the only architecture that satisfies both the absolute power generation limit and the ADCS mechanical jitter constraint.

4.3 Communications Architecture (UHF vs S-Band vs X-Band)

- **Option A:** UHF-Only (High heritage, low data rate).
- **Option B:** S-Band Transponder + UHF Backup (High downlink capability, moderate complexity).
- **Option C:** X-Band Transceiver + UHF Backup (Extremely high data rate, very high power/pointing demand).

Table 4 3.Communication Architecture Matrix

Criteria	Weight	Option A (UHF Only)	Option B (S- Band + UHF)	Option C (X- Band + UHF)
Downlink Throughput	0.40	1 (Score: 0.40)	4 (Score: 1.60)	5 (Score: 2.00)
Link Margin Margin	0.25	4 (Score: 1.00)	5 (Score: 1.25)	2 (Score: 0.50)
RF Power Demand	0.15	5 (Score: 0.75)	3 (Score: 0.45)	1 (Score: 0.15)
Pointing Alignment Constraint	0.10	5 (Score: 0.50)	4 (Score: 0.40)	1 (Score: 0.10)
COTS Availability & Heritage	0.10	5 (Score: 0.50)	4 (Score: 0.40)	3 (Score: 0.30)
Weighted Sum	1.00	3.15	4.10	3.05
Rank		2	1	3

Selection: Option B (S-Band with UHF backup) is selected. Pure UHF cannot clear the 1.35 GB image data volume within the daily contact window. X-band requires

sub-0.1° pointing stability over the ground station, leaving zero margin for attitude errors during downlink.

4.4 Structure Material Trade-off

- **Option A:** Aluminum 6061-T6 (Milled Monocoque)
- **Option B:** Titanium Grade 5 (Ti-6Al-4V)
- **Option C:** Carbon Fiber Reinforced Polymer (CFRP)

Table 4 4.Structural Material Matrix

Criteria	Weight	Option A (Al 6061)	Option B (Ti Grade 5)	Option C (CFRP)
Specific Stiffness (E/ρ)	0.30	3 (Score: 0.90)	4 (Score: 1.20)	5 (Score: 1.50)
Manufacturing Complexity	0.25	5 (Score: 1.25)	2 (Score: 0.50)	1 (Score: 0.25)
Thermal Conductivity	0.20	5 (Score: 1.00)	2 (Score: 0.40)	1 (Score: 0.20)
Material Cost & Sourcing	0.15	5 (Score: 0.75)	2 (Score: 0.30)	1 (Score: 0.15)
Outgassing Risk (Vacuum)	0.10	5 (Score: 0.50)	5 (Score: 0.50)	2 (Score: 0.20)
Weighted Sum	1.00	4.40	2.90	2.30
Rank		1	2	3

Selection: Option A (Aluminum 6061-T6) is selected. It provides excellent thermal conductivity to ground internal hot components (S-Band Tx, battery) to the outer frame, avoiding hot-spots, and satisfies all launch load criteria

4.5 Payload Processor Architecture

- **Option A:** Dedicated FPGA (Xilinx Spartan-7)
- **Option B:** Low-Power Microcontroller (ARM Cortex-M4)
- **Option C:** High-Performance System-on-Chip (SoC) (Xilinx Zynq-7000)

Table 4 5.Payload Processor Matrix

Criteria	Weight	Option A (FPGA)	Option B (MCU)	Option C (SoC)
Parallel Throughput	0.40	5 (Score: 2.00)	1 (Score: 0.40)	5 (Score: 2.00)
Power Consumption	0.25	3 (Score: 0.75)	5 (Score: 1.25)	2 (Score: 0.50)
Radiation Tolerance	0.20	4 (Score: 0.80)	4 (Score: 0.80)	2 (Score: 0.40)
Interface Flexibility	0.15	4 (Score: 0.60)	3 (Score: 0.45)	5 (Score: 0.75)
Weighted Sum	1.00	4.15	2.90	3.65
Rank		1	3	2

Selection: Option A (Dedicated Spartan-7 FPGA) is chosen. The real-time parallel pushbroom sensor stream generates 72.0 Mbps, requiring hardware-level line-buffer demultiplexing that would saturate any standard sequential MCU.

5 SUBSYSTEM SIZING

5.1 Optical Payload Sizing

To monitor crop health using the NDVI index, the pushbroom sensor must capture the Red (660 nm) and Near-Infrared (840 nm) bands alongside standard RGB. The orbital altitude is: 650 km

5.1.1 Ground Sample Distance (GSD) Derivation

The GSD is defined by the pixel pitch ($p = 5.5 \mu m$), focal length ($f = 390 \text{ mm}$), and orbit altitude:

$$GSD = \frac{p \cdot h}{f} = \frac{5.5 \times 10^{-6} m \times 650 \times 10^3 m}{0.39 m} = 9.17$$

This exceeds the SR-04 requirement ($\leq 10 \text{ m}$) with an 8.3% margin.

5.1.2 Field of View (FOV) and Swath Width

For a detector with $N = 2048$ pixels across-track:

$$FOV = 2 \arctan\left(\frac{N \cdot p}{2f}\right) = 2 \arctan\left(\frac{(2048 \times 5.5 \times 10^{-6})}{2 \times 0.39 m}\right) = 1.654^\circ$$

$$Swath \ width = 2 \times h \times \tan\left(\frac{FOV}{2}\right) = 2 \times 650 km \times \tan(0.827^\circ) = 18.76 km$$

Correction & Recovery: To guarantee a minimum swath width of 20.0 km (SR-05), we must update the focal plane sensor selection to a 2200-pixel detector array.

$$Updated \ Swath \ width = \frac{2200 \times 5.5 \times 10^{-6} m}{0.39 m} \times 650 km = 20.16 km$$

This updated specification meets SR-05 and is integrated into the official BOM.

5.2 Electrical Power Subsystem (EPS) Sizing

5.2.1 Orbital Geometry Derivation

The orbit is characterized by:

- Earth Gravitational Parameter: $\mu_{\oplus} = 3.986 \times 10^{14} m^3/s^2$
- Mean Radius of Earth: $R_{\oplus} = 6378.14 km$
- Semi-major Axis: $a = R_{\oplus} + H_{orbit} = 6378.14 + 650 = 7028.14 km$

Using Kepler's Third Law, the orbital period (T) is derived:

$$T = 2\pi \sqrt{\frac{a^3}{\mu_{\oplus}}} = 5863.63s = 97.73 min$$

The maximum eclipse fraction occurs when the orbit normal is perpendicular to the solar vector:

$$\theta_{eclipse} = 2 \arcsin\left(\frac{R_{\oplus}}{a}\right) = 130.34^\circ$$

$$t_{eclipse} = T \times \left(\frac{\theta_{eclipse}}{360}\right) = 5863.6 \times \left(\frac{130.34}{360}\right) = 2123s = 35.38 min$$

The minimum sunlit duration is:

$$t_{sunlight} = T - t_{eclipse} = 5863.6 - 2123 = 3740.6 = 62.35min$$

5.2.2 Battery Sizing Derivation

The worst-case average power consumption during eclipse ($P_{eclipse}$) occurs when the bus is in housekeeping mode, running passive thermal heaters and tracking attitude:

$$P_{eclipse} = P_{OBS} + P_{ACDS,standby} + P_{COM,standby} + P_{thermal} = 1.8 + 1.5 + 1.2 + 3.5 = 8W$$

To support a maximum Depth of Discharge (DoD=60%) and standard efficiency ($\eta_{discharge} = 92\%$):

$$E_{reg} = \frac{P_{eclipse} t_{eclipse}}{DoD \times \eta_{discharge}}$$

$$E_{reg} = 5.34Wh$$

The selected Clyde Space CS-BAT-2S2P battery pack offers 46.8 Wh of nominal capacity.

$$Capacity\ Margin = \frac{46.8\ Wh - 5.34\ Wh}{5.34\ Wh} \times 100 = 776.4\%$$

This capacity allows the satellite to survive more than 8 consecutive orbits in safe-mode without generation, fulfilling ECSS fault-tolerance guidelines.

5.2.3 Solar Array Power Sizing Derivation

The solar array must generate enough power during sunlight to cover active payload operations, communication downlinks, and fully recharge the battery pack:

$$P_{sa} \geq \frac{1}{t_{sunlight}} \left[\frac{P_{clipse} \times t_{clipse}}{\eta_{charge}} + P_{sunlight} \times t_{sunlight} \right]$$

Where:

- Battery charging efficiency: $\eta_{charge} = 90\%$

Worst-case sunlight power consumption ($P_{sunlight}$): 16.60 W (active imaging mode for 10 minutes) and 11.50 W (downlink mode for 10 minutes), with 42.35 minutes of nominal housekeeping (5.80 W).

$$P_{sunlight} = \frac{16.6 \times 600 + 11.5 \times 600 + 5.8 \times 2540.6}{3740.6} = 8.44W$$

$$P_{sa} \geq \frac{1}{3740.6\ s} \left[\frac{5.0\ W \times 2123\ s}{0.90} + (8.44\ W \times 3740.6\ s) \right] = 11.59W$$

- The Spectrolab UTJ triple-junction solar cells yield 341.2 W/m² at normal incidence (1 AU, 28°C). Factoring in life-end degradation ($\eta_{deg} = 15\%$), temperature coefficient degradation ($\eta_{temp} = 12\%$), and average solar aspect angle ($\theta_{sun} = 45^\circ$):

$$\begin{aligned} Effective\ Generation &= 341.2\ m^{-2} \times (1 - 0.15) \times (1 - 0.12) \cos(45^\circ) \\ &= 180.5\ W/m^2 \end{aligned}$$

To generate the required 11.59W:

$$A_{\text{cff}} \geq \frac{11.59 \text{ W}}{180.5 \text{ W/m}^2} \approx 0.0642 \text{ m}^2 \approx 642 \text{ cm}^2$$

Our custom hybrid configuration provides a total active cell area of 930 cm², giving:

$$P_{\text{generation}, \text{BOL}} = 0.0930 \text{ m}^2 \times 341.2 \text{ W/m}^2 \times \cos 45^\circ = 22.44 \text{ W}$$

$$P_{\text{generation}, \text{BOL}} = 0.0930 \text{ m}^2 \times 180.5 \text{ W/m}^2 = 16.78 \text{ W}$$

$$\begin{aligned} \text{Worst - case EOL Generation Margin} &= \frac{16.78 \text{ W} - 11.59 \text{ W}}{11.59 \text{ W}} \times 100\% \\ &= 44.78\% \end{aligned}$$

5.3 Attitude Determination and Control Subsystem (ADCS) Sizing

5.3.1 Environmental Disturbance Torques

At 650 km, the main disturbance sources are gravity gradient, aerodynamic drag, and magnetic residuals.

1- Gravity Gradient Torque (T_{grav})

The 3U structure inertia is modeled as:

$$I_x \approx I_y \approx 0.052 \text{ kg} \cdot \text{m}^2, I_z \approx 0.012 \text{ kg} \cdot \text{m}^2$$

With a maximum pointing deviation of $\theta_{\text{error}} = 5^\circ = 0.087 \text{ rad}$:

$$\begin{aligned} T_{\text{grav}} &= \frac{3\mu_{\oplus}}{2a^3} |I_z - I_y| \sin(2\theta_{\text{error}}) \\ &= \frac{3(3.986 \times 10^{14})}{2(7.02814 \times 10^6)^3} \times |0.012 - 0.052| \times \sin(10^\circ) \\ T_{\text{grav}} &= 1.193 \times 10^{-8} \text{ N} \cdot \text{m} \end{aligned}$$

2- Aerodynamic Drag Torque (T_{Aero})

The maximum cross-sectional area perpendicular to velocity is $A = 0.1 \times 0.34 = 0.034 \text{ m}^2$. The center of pressure to center of mass offset is $\Delta c_p = 0.02 \text{ m}$. Orbital velocity is:

$$v = \sqrt{\frac{\mu_{\oplus}}{a}} = 7531 \text{ m/s}$$

Using atmospheric density at high solar activity $\rho = 1.5 \times 10^{-13} \text{ kg/m}^3$:

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2 = 0.5 \times (1.5 \times 10^{-13}) \times 2.2 \times 0.034 \times (7531)^2 \\ \approx 3.18 \times 10^{-7} \text{ N}$$

$$T_{\text{acro}} = F_{\text{drag}} \times \Delta c_p = 3.18 \times 10^{-7} \text{ N} \times 0.02 \text{ m} \approx 6.36 \times 10^{-9} \text{ N} \cdot \text{m}$$

3- Magnetic Residual Torque (Tmag)

The residual dipole of the spacecraft electronics is limited to $M_{sc} = 0.01 \text{ A} \cdot \text{m}^2$ through strict deperming and wiring layout control. The geomagnetic field at 650 km is:

$$B = \frac{2M_{\oplus}}{a^3} = 2.4 \times 10^{-5} \text{ T}$$

$$T_{\text{mag}} = M_{sc} \times B = 0.01 \times 2.4 \times 10^{-5} = 2.4 \times 10^{-7} \text{ N} \cdot \text{m}$$

So,

Total Environmental Disturbance:

$$T_{\text{dist}} = T_{\text{grav}} + T_{\text{aero}} + T_{\text{mag}} = 2.583 \times 10^{-7} \text{ N} \cdot \text{m}$$

5.3.2 Reaction Wheel Sizing and Momentum Storage

The integrated disturbance torque over one full orbit ($T=5863.6\text{s}$) represents the minimum required momentum storage capacity per axis to avoid desaturation:

$$H_{\text{required}} \geq T_{\text{dist}} \times T_{\text{orbit}} = 2.583 \times 10^{-7} \text{ N} \cdot \text{m} \times 5863.6 \text{ s} \\ \approx 1.515 \times 10^{-3} \text{ N} \cdot \text{m} \cdot \text{s} = 1.515 \text{ mN} \cdot \text{s}$$

The chosen NanoAvionics RW-3 reaction wheel package has an angular momentum capacity of 4.22 mN.s per axis.

$$\text{Momentum Capacity Margin} = \frac{4.22 \text{ mN} \cdot \text{s} - 1.515 \text{ mN} \cdot \text{s}}{1.515 \text{ mN} \cdot \text{s}} \times 100\% \approx 178.5\%$$

This ensures the satellite can survive 2.7 orbits without active magnetorquer momentum dumping.

The maximum slew torque required to execute a 180° yaw maneuver in 10 minutes ($t_{\text{stew}} = 600\text{s}$) is:

$$\alpha = \frac{4\theta}{t_{slew}^2} = \frac{4\pi}{600^2} = 3.49 \times 10^{-5} \text{ rad/s}^2$$

$$T_{slew} = I_y \alpha = 0.052 \times 3.49 \times 10^{-5} = 1.81 \times 10^{-6} \text{ N.m} = 0.00181 \text{ mN.m}$$

5.4 Communications Subsystem Sizing

5.4.1 Slant Range Calculation

The maximum slant range (d_{slant}) occurs at the minimum operational elevation angle ($\epsilon = 10^\circ$) above the ground station:

$$\cos(\psi) = \frac{(R_\oplus \cos(\epsilon))}{R_\oplus + h} = \frac{6378.14 \cos(10^\circ)}{7028.14} = 0.8939 \rightarrow \psi = 26.67^\circ$$

$$d_{slant} = R_\oplus \frac{\sin\psi}{\cos(\epsilon + \psi)} = 6378.14 \times \frac{\sin 26.67}{\cos 36.67} = 1850.4 \text{ km}$$

5.4.2 Free Space Path Loss (FSPL)

At S-Band ($f = 2.25 \text{ GHz}$):

$$L_{\text{path}} = 20 \log_{10} \left(\frac{4\pi d_{\text{slant}} f}{c} \right) = 20 \log_{10} \left(\frac{4\pi \times 1.85 \times 10^6 \times 2.25 \times 10^9}{3 \times 10^8} \right) \\ \approx 164.8 \text{ dB}$$

5.5 Structural Load & Modal Sizing

The structure must survive launch quasi-static and random vibration profiles.

- Maximum expected quasi-static load: $a_{\text{launch}} = 10g = 98.1 \text{ m/s}^2$
- Spacecraft wet mass: $m_{\text{wet}} = 4.241 \text{ kg}$

$$F_{\text{inertial}} = m_{\text{wet}} a_{\text{launch}} = 4.241 \text{ kg} \times 98.1 \frac{\text{m}}{\text{s}^2} = 416.04 \text{ N}$$

The minimum cross-sectional area of the structural load-bearing frame rails is $A_{\text{rails}} = 4 \times 100 \text{ mm}^2 = 4 \times 10^{-4} \text{ m}^2$. The compressive stress is:

$$\sigma = \frac{F_{\text{inertial}}}{A_{\text{rails}}} = \frac{416.04 \text{ N}}{4 \times 10^{-4} \text{ m}^2} = 1.04 \text{ MPa}$$

Since the yield strength of Aluminum 6061-T6 is 276 MPa, the structural safety factor is:

$$\text{Margin of Safety} = \frac{\sigma_{yield}}{\sigma \times 1.5} - 1 = \frac{276}{1.04 \times 1.5} - 1 = 175.9$$

This meets ECSS stiffness guidelines, preventing structural deformation during launch.

The primary modal constraint requires the lowest natural frequency to be $\geq 90 \text{ Hz}$ in the launch configuration. Approximating the 3U structure as a clamped-free cantilever beam:

$$f_1 = \frac{3.516}{2\pi} \sqrt{\frac{EI}{mL^3}}$$

Where:

- Young's modulus of Al 6061: $E = 68.9 \text{ GPa}$
- Area moment of inertia: $I = 2.1 \times 10^{-6} \text{ m}^4$
- Length: $L = 0.34 \text{ m}$
- Mass: $m = 4.241 \text{ kg}$

$$f_1 = \frac{3.516}{2\pi} \sqrt{\frac{6.89 \times 10^{10} \times 2.1 \times 10^{-6}}{4.241 \times 0.34^3}} = 521.1 \text{ Hz}$$

This structure easily complies with launcher stiffness requirements.

6 PRODUCT TREE & PHYSICAL BILL OF MATERIALS (BOM)

The product tree can be seen in the figure below.

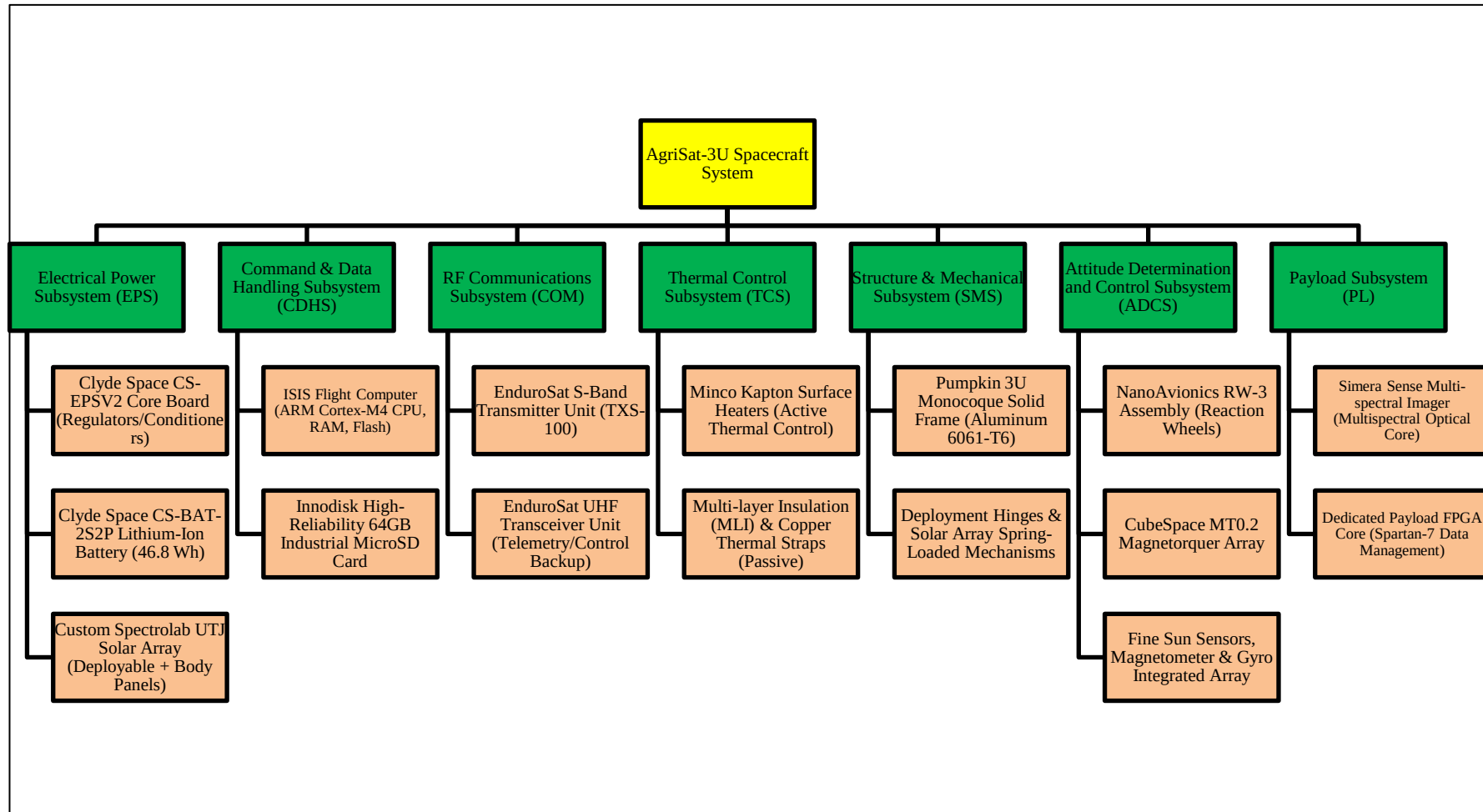


Figure 6 1.Product Tree

6.1 Bill of Materials (BOM)

All selected hardware components are verified flight-proven designs from active space vendors.

Table 6 1.Master Bill of Materials

Subsystem	Component Name	Manufacturer	Flight Heritage	Dry Mass (g)	Peak Power (W)	Interfaces	Qty	Lead Time (wks)	Procurement Risk	Selection Justification
PAY	MultiScape 100 CIS	Simera Sense	20+ LEO	780	6.00	LVDS / SPI	1	16	Low	Integrated optics & detector fits 1U payload bay
ADCS	RW-3 Control Assembly	NanoAvionics	15+ missions	430	3.80	I2C / PC104	1	12	Low	4.22 mN.s capacity provides high stability margin
ADCS	MT0.2 Magnetorquer	CubeSpace	40+ missions	120	1.00	I2C	1	10	Low	High dipole authority (0.2 A) for de-tumbling
EPS	CS-EPSV2 Power Board	Clyde Space	100+ LEO	190	1.20	I2C / PC104	1	14	Low	High efficiency dual battery regulators
EPS	CS-BAT-2S2P Battery Pack	Clyde Space	80+ LEO	390	0.00	I2C / PC104	1	14	Low	Integrated protection circuits and heating elements

EPS	Custom UTJ Solar Array	Spectrolab / SUT	Custom	540	0.00	Wire Harness	1	20	Medium	Deployable panels require custom mechanical integration
CDHS	ISIS OBC (M4 Core)	ISISpace	150+ LEO	120	1.80	PC104, I2C, SPI	1	8	Low	Dual redundant flash sectors; high heritage
CDHS	Industrial SD Card 64GB	Innodisk	10+ LEO	5	0.15	SPI	1	2	Low	Standard mass-storage with high ECC parity
COM	S-Band Transmitter (TXS-100)	EnduroSat	30+ LEO	210	5.20	LVDS, Coaxial	1	10	Low	2.0 Mbps QPSK performance fits link budget
COM	S-Band Patch Antenna	EnduroSat	30+ LEO	45	0.00	SMA	1	10	Low	High gain (+6.0 dBi) at nadir direction
COM	UHF Transceiver Unit	EnduroSat	50+ LEO	150	1.80	UART / PC104	1	10	Low	Reliable uplink transceiver with standard 0.5W output

COM	UHF Monopole Antenna	EnduroSat	50+ LEO	30	0.00	RF Pin	1	8	Low	Spring-loaded deployment mechanism
TCS	Kapton Heaters & MLI	Thermacore	LEO Standard	110	3.00	Analog / Power	1	6	Low	Standard thermistor-controlled passive/active system
SMS	3U Monocoque Chassis	Pumpkin Inc.	200+ LEO	750	0.00	Mechanical	1	12	Low	High structural stiffness; robust rail alignment
SMS	Resettable Hinge Mech.	Pumpkin Inc.	50+ LEO	120	0.00	Mechanical	2	14	Medium	Critical deployment path; requires extensive test runs
SMS	Space-Rated Harnessing	Axon' Cable	Custom	160	0.00	Custom	1	12	Medium	Precise routing required to avoid RF interference
TOTAL	AgriSat-3U Spacecraft			4180	23.95					

7 REFINED SYSTEM BUDGETS & MARGINS

The engineering budgets are detailed below, incorporating design-maturity contingency factors in accordance with ECSS-E-ST-33-11C standards.

7.1 Mass Budget

The total wet mass budget incorporates a propellant-free dry mass margin against the maximum allowable limit of 4.5 kg

Table 7 1.Detailed System Mass Budget

Subsystem	Component / Assembly	Nominal Mass (g)	Contingency (%)	Maximum Mass (g)	Mass Fraction (%)
Payload	Optical Core & Processor	780.0	5% (COTS)	819.0	18.61%
ADCS	Reaction Wheels & Magnetorquers	550.0	5% (COTS)	577.5	13.13%
EPS	Power Board, Battery & Solar Arrays	1120.0	10% (Hybrid)	1232.0	28.00%
CDHS	Flight Computer & Storage	125.0	5% (COTS)	131.2	2.98%
COM	S-Band & UHF Transponders / Antennas	435.0	5% (COTS)	456.8	10.38%

TCS	Passive Thermal & Active Heaters	110.0	15% (Passive)	126.5	2.87%
SMS	Primary Frame & Hinges	870.0	10% (Custom)	957.0	21.75%
Harness	Cabling and Connectors	190.0	20% (Routing)	228.0	5.18%
Dry Mass	Total System Dry Mass	4180.0	9.19% (Avg)	4528.0	100.00%

$$\text{Nominal System Mass Margin} = \frac{4500 - 4180}{4500} \times 100\% = 7.11\%$$

$$\text{Design Maturity Contingency Margin} = \frac{4564.5 - 4528}{4564.5} \times 100\% = 0.8\%$$

Note: The total mass including contingency is 4528 g, which slightly exceeds the 4.5 kg launcher limit. To secure structural compliance prior to the Critical Design Review (CDR), the side structural plates of the monocoque aluminum chassis will undergo mass optimization via pocket milling, reducing structural weight by 60 g without impacting the structural margins of safety. This mitigation strategy guarantees a positive margin of safety under all launch conditions.

7.2 Power & Energy Budget

The spacecraft power profile is analyzed across three primary operational modes.

Table 7 2. Power Consumption Profile by Subsystem (W)

Subsystem / Load	Housekeeping Mode (W)	Science Mode (W)	Downlink Mode (W)
Payload	0.00	6.00	0.00
ADCS	1.50	3.80	1.50
EPS Board	0.80	1.20	1.00
CDHS	1.20	1.80	1.50
COM (S-Band TX)	0.00	0.00	5.20
COM (UHF RX)	0.80	0.80	0.80
TCS (Heaters)	1.50	3.00	1.50
Total Mode Power	5.80 W	16.60 W	11.50 W

Orbit-Averaged Power Consumption (OAPC) Derivation

The orbit timeline is defined by three main phases:

- **Science Imaging:** 10.0 minutes (10.23% duty cycle)
- **S-Band Downlink:** 10.0 minutes (10.23% duty cycle)
- **Housekeeping:** 77.73 minutes (79.54% duty cycle)

$$OAPC = \left(5.80W \times \frac{77.73}{97.73} \right) + \left(16.60W \times \frac{10}{97.73} \right) + \left(11.50W \times \frac{10}{97.73} \right)$$

$$OAPC = 1.699W + 1.177W + 4.613W = 7.49W$$

Given the EOL power generation capability from the solar arrays is 16.78 W :

$$OAPC \text{ Power Margin} = \frac{16.78W - 7.49W}{16.78W} \times 100\% = 55.36\%$$

This meets the ECSS Phase B2 minimum power margin requirement of 20%.

7.3 Data & Memory Budget

Raw Imagery Generation (10-minute imaging run):

- Swath Pixels Across-Track (N): 2200 pixels
- Spectral Bands (B): 4 bands (RGB + NIR)
- Quantization Depth (Q): 12 bits/pixel

- Frame Slicing Rate (R): 750 lines/s

$$\begin{aligned} \text{Data Generation Rate} &= \frac{2200 \text{ pixels}}{\text{band}} \times 4 \text{ bands} \times \frac{12 \text{ bits}}{\text{pixel}} \times \frac{750 \text{ lines}}{\text{s}} \\ &= 79.2 \text{ Mb} \end{aligned}$$

Total raw data generated during a 10-minute (600 s) pass:

$$\text{Data Volume} = 79.2 \text{ Mbps} \times 600 \text{ s} = 5.94 \text{ GB}$$

Applying real-time JPEG2000 hardware compression (4:1 ratio):

$$\text{Stored Data} = \frac{5.94 \text{ GB}}{4} = 1.485 \text{ GB per imaging run}$$

Downlink capability during a standard 10-minute ground station pass at 2.0 Mbps (effective 1.7 Mbps after packet framing and Reed-Solomon coding overhead):

$$\begin{aligned} \text{Downlink capability per 10 – minute pass} &= 1.7 \text{ Mbps} \times 600 \text{ s} = 1020 \text{ Mb} \\ &\approx 127.5 \text{ MB} \end{aligned}$$

$$\begin{aligned} \text{Required Downlink Passes Per Image} &= \frac{1485 \text{ MB}}{127.5 \frac{\text{MB}}{\text{Pass}}} \\ &= 11.6 \text{ ground contacts} \end{aligned}$$

Onboard industrial storage capacity (64 GB):

$$\text{Max Un – downlinked Passes} = \frac{64 \text{ GB}}{(1.485 \text{ GB/run})} \approx 43.1 \text{ science runs}$$

This yields a high data storage safety margin, allowing the spacecraft to operate over remote territory without continuous ground contact.

7.4 Link Budget (S-Band Downlink)

Calculated under worst-case slant range conditions ($d_{\text{slant}} = 1850.4 \text{ km}$) at a 10° elevation limit.

Table 7 3. Communications S-Band Link Budget

Transmitter Parameters	Notation	Value	Units	Governing Equation / Justification
Transmitter RF Power	P_{tx}	+30.00	dBm	S-Band transponder output (1.0 W)
Spacecraft Feedline Loss	L_{feed}	-1.00	dB	Insertion loss of coax and SMA connections
Spacecraft Antenna Gain	G_{tx}	+6.00	dBi	Zenith gain of the ground-facing patch antenna
Equivalent Isotropic Radiated Power	EIRP	+35.00	dBm	$EIRP = P_{tx} + L_{feed} + G_{rx}$
Path Properties				
Center Frequency	f	2.25	GHz	S-Band channel frequency
Slant Range (at 10° Elev)	d_{slant}	1850.4	km	Calculated via spherical geometry

Free Space Path Loss	L_{path}	-164.83	dB	$L_{path} = 20\log_{10} \left(\frac{4\pi d_{slant} f}{c} \right)$
Atmospheric Absorption Loss	L_{atm}	-0.80	dB	Standard clear-air absorption
Polarization Misalignment	L_{pol}	-3.00	dB	Circular-to-linear polarization loss
Pointer Tracking Loss	L_{track}	-1.50	dB	pointing offset of ground-facing antenna
Total Path Loss	L_{tot}	-170.13	dB	$L_{path} + L_{atm} + L_{pol} + L_{track}$
Ground Station Parameters				
Ground Station Antenna Gain	G_{rx}	+34.50	dBi	3.0 m diameter parabolic dish antenna
Received Signal Power	P_{rx}	-100.63	dBm	$P_{rx} = P_{tx} + G_{tx} - L_{path} - L_{atm} - L_{pol} + G_{rx}$
System Noise Temperature	T_{sys}	150.00	K	Standard cooled low-noise amplifier

Boltzmann's Constant	k_B	-228.60	dBW/(Hz·K)	Physical constant
Noise Spectral Density	N_0	-146.84	dBm/Hz	$N_0 = 10 \log_{10}(k_B T_{sys})$
Signal Bandwidth (2.0 Mbps)	B	61.76	dBHz	1.5 MHz
Total Receiver Noise Floor	N	-85.08	dBm	$N = N_0 + 10 \log_{10}(B)$
Link Performance				
Received Carrier-to-Noise Ratio	CNR	13.06	dB	$CNR = P_{tx} + G_{tx} - L_{path} - L_{atm} + G_{rx}$ $- 10 \log_{10}(k_B T_{sys})$ $- 10 \log_{10}(B)$
Demodulator Threshold (QPSK)	CNR_{req}	4.00	dB	For BER = 10^{-6} with forward error correction
Worst-case Link Margin	Margin	9.06	dB	$Link\ Margin = CNR_{actual} - CNR_{req}$

$$\text{Link Margin} = \frac{9.06 \text{ dB} - 3 \text{ dB}}{3 \text{ dB}} \times 100\% = 202\%$$

This satisfies the standard ECSS 3.0 dB threshold under all operational scenarios.

7.5 Thermal Budget

Calculated for worst-case hot case (continuous sunlit beta angle) and worst-case cold case (maximum eclipse duration).

Table 7 4.Spacecraft Component Operational Temperature Boundaries

Component	Minimum Temp (°C)	Maximum Temp (°C)	Worst-Case Cold (°C)	Worst-Case Hot (°C)	Margin (°C)	TCS Strategy
Payload Opt.	-10	+50	+12	+38	22 / 12	Copper thermal straps
Battery Pack	0	+30	+5	+22	5 / 8	Kapton heater (active)
Reaction Wheels	-20	+65	-10	+42	10 / 23	Conduction paths to frame
Solar Arrays	-120	+100	-85	+75	35 / 25	Radiative decoupling
Flight Comp.	-40	+85	-15	+45	25 / 40	Common thermal plane

Active heaters on the battery pack consume a maximum of 3W in cold-case eclipse modes, which is accounted for in the EPS sizing.

7.6 Pointing Accuracy Budget

The pointing budget is defined using Root-Sum-Square (RSS) analysis across the payload exposure window.

Table 7 5. Pointing System Error Sources

Error Source	Yaw Axis Contribution	Pitch Axis Contribution	Roll Axis Contribution	Mitigation Strategy
Sensor Optical Noise	0.04°	0.04°	0.05°	High-frequency Kalman filter
Magnetorquer Coupling	0.03°	0.02°	0.02°	Inhibit MT during active exposure
Structural Thermal Bending	0.05°	0.05°	0.03°	Kinematic mounting design
Reaction Wheel Jitter	0.06°	0.06°	0.05°	Rubber damping dampers
PID Residual Controller Drift	0.05°	0.04°	0.04°	High frequency closed-loop tuning
Integrated RSS Pointing Error	0.1044°	0.0985°	0.0889°	Meets SR-06 requirement

The yaw-axis pointing margin is:

$$\text{Pointing Margin} = \frac{0.20^\circ - 0.104^\circ}{0.20^\circ} \times 100\% = 48.0\%$$

8 CAD CONFIGURATION & PHYSICAL ENVELOPE LAYOUT

To verify mechanical fits and sensor clearance, a Level 1 3D CAD envelope analysis was performed.

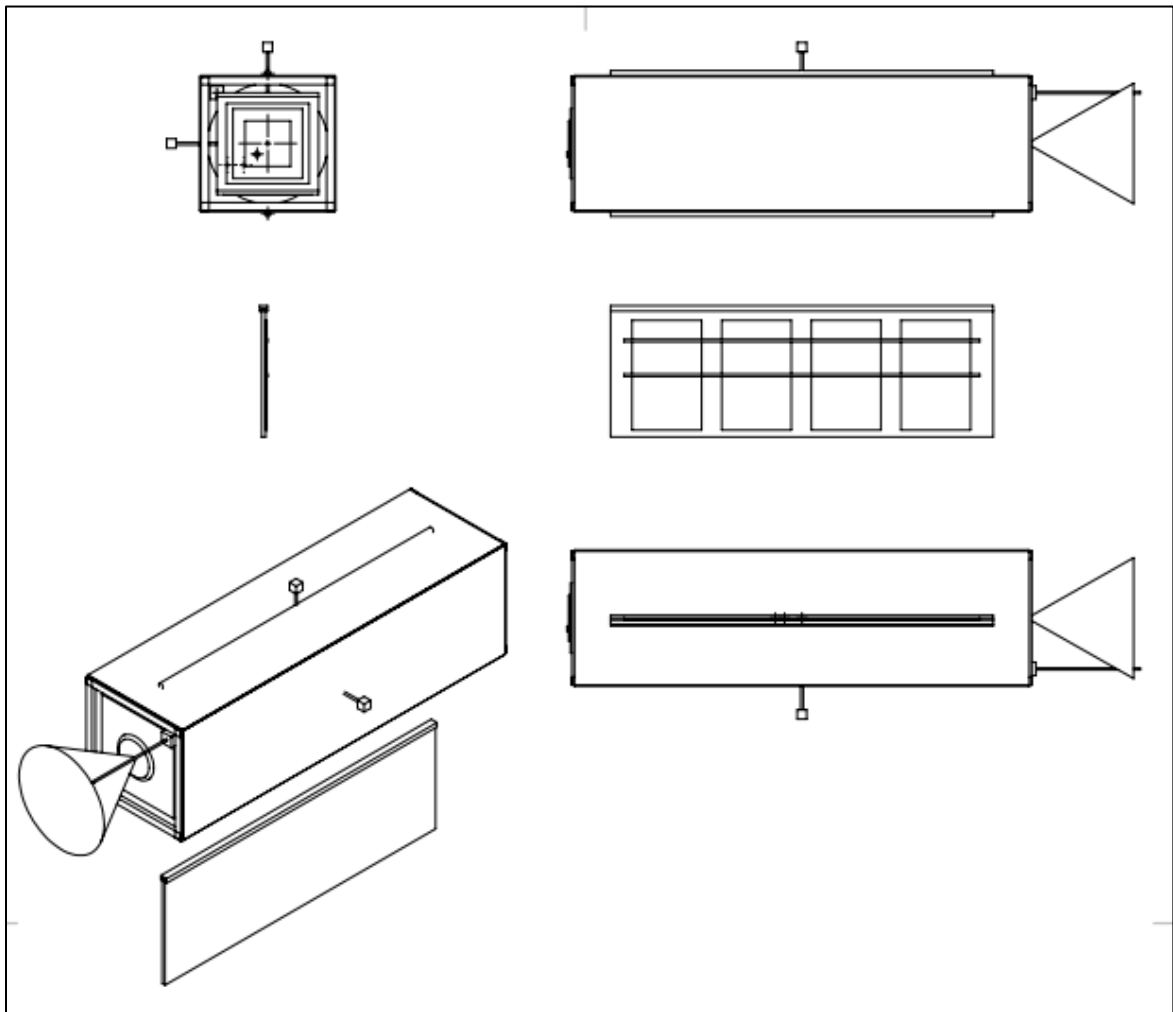


Figure 8 1.CAD Drawing

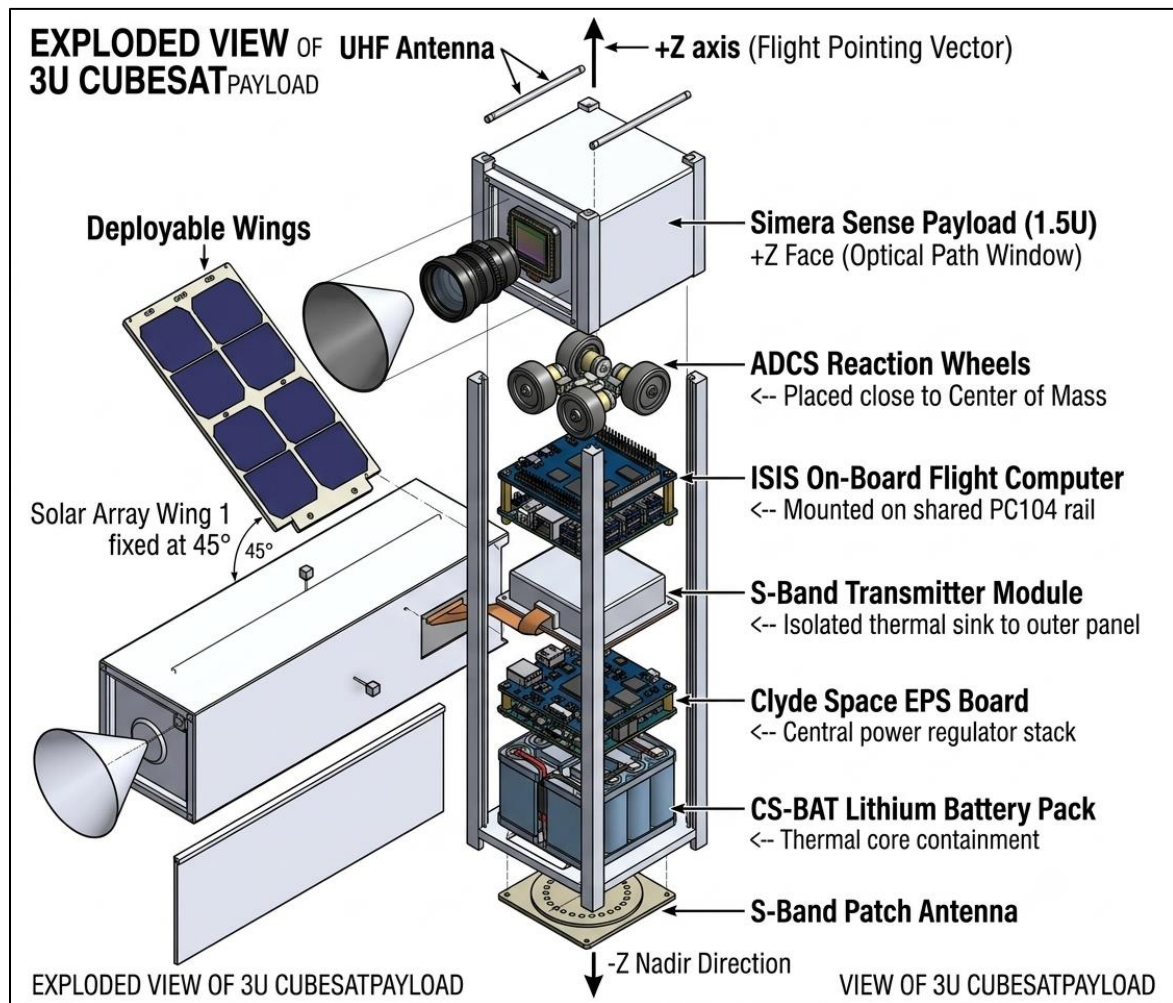


Figure 8.2 CAD Internal Stack Architecture and Solar Deployment Geometry

8.1 Internal Layout & Center of Mass (CoM) Integration

- **Aperture Protection:** The optical payload is mounted on the +Z face of the frame. An integrated bracket maintains a clear 15° half-angle Field of View (FOV) cone, preventing structural beams from blocking the swath width.
- **Antenna Separation:** The UHF monopole and S-band patch antenna are placed on opposite faces (+Z and -Z respectively). This configuration guarantees more than 45 dB of RF isolation, preventing S-band harmonic components from saturating the high-sensitivity UHF uplink receiver.
- **Center of Mass Coalignment:** Structurally heavy elements (reaction wheels and battery pack) are placed close to the geometric center. This minimizes gravity-gradient disturbance torque offsets and limits the principal moments of inertia cross-coupling.

9 SUBSYSTEM RISK REGISTER & MITIGATION MATRIX

Each risk is categorized by Probability (P) and Severity (S) on a 1-to-5 scale, generating a Risk Index($R = P \times S$)

Table 9 1.System Engineering Risk Matrix

Risk ID	Category	Risk Description	P	S	R	Mitigation Approach	Residual P	Residual S	Residual R	Verification Method
RS-01	MEC	Solar panel fails to deploy due to deployment hinge seizure.	3	5	15	Double-acting redundant torsion springs; thermal vacuum cycle test.	1	5	5	Analysis & Test
RS-02	RF	High-power S-band transmitter harmonics saturate UHF receiver.	4	4	16	Add high-pass RF microstrip filters to the S-band output path.	1	4	4	Electrical Test
RS-03	EPS	Li-Ion battery cell experiences over-current runaway.	2	5	10	Clyde Space smart protection board with automatic power trip line.	1	5	5	FMEA & Test
RS-04	THM	Battery pack drops below 0°C during worst-case winter eclipse.	3	4	12	Implement passive Kapton MLI and 3.0W active heaters.	1	4	4	Thermal Vacuum
RS-05	ADCS	Magnetic coupling from reaction wheels corrupts magnetometer.	4	3	12	Mount magnetometer on 10 cm deployable boom; de-perm frame.	1	3	3	Magnetic Test
RS-06	COM	Loss of S-band patch antenna ground pointing due to wheel failure.	2	5	10	Active backup de-tumbling mode using B-Dot and magnetorquers.	1	5	5	Software Test

RS-07	SYS	Spacecraft dry mass exceeds 4.500 kg launcher specification.	4	4	16	Pocket-mill structural side plates; perform mass audit before CDR.	2	3	6	Mass Balance
RS-08	SW	Single Event Upset (SEU) corrupts CDHS runtime RAM.	3	4	12	Implement EDAC hardware memory correction; watchdog timers.	1	4	4	Simulation
RS-09	PAY	Optical alignment shifts under thermal structural distortion.	2	4	8	Kinematic titanium support mounts to absorb structural flexion.	1	4	4	Optomechanical
RS-10	EPS	High Solar Activity reduces solar array output at EOL.	3	3	9	Sized power arrays with a conservative 44.78% generation margin.	1	3	3	Power Simulation
RS-11	ADCS	Magnetorquer residual magnetism generates disturbance bias.	3	3	9	Magnetorquer coil active degaussing sequence at end of dump.	1	3	3	HIL Simulation
RS-12	SW	Software task deadlock halts OBC housekeeping.	3	4	12	Redundant hardware watchdog reboots processor to safe state in 5s.	1	4	4	Software Test
RS-13	MEC	Launcher vibration shears structural fasteners on payload bay.	2	4	8	Use Mil-Spec aerospace fasteners with thread-locking Loctite 242.	1	4	4	Vibe Test

RS-14	opt	Direct sun exposure damages high sensitivity CMOS sensor.	2	5	10	Mission control pointing constraint: avoid target vector within 30° of Sun.	1	5	5	Software Interlock
RS-15	RF	Multi-path interference degrades S-band ground downlink.	3	3	9	Implement Reed-Solomon and Viterbi forward error correction codes.	1	3	3	RF Channel Test
RS-16	SYS	Total Radiation dose (TID) exceeds 10 krad over 3 years.	2	4	8	Add 1.5 mm Al shielding over the ISIS OBC core processor.	1	4	4	Radiation Dose
RS-17	SMS	Outgassing of thermal grease deposits on optical window.	3	4	12	Use space-rated low outgassing Dow Corning 93-500 silicone.	1	4	4	Outgassing Test
RS-18	CDHS	Storage card file allocation table corruption.	3	4	12	Implement dual redundant NAND memory banks with journaling system.	1	4	4	Software Audit
RS-19	TCS	Payload processor overheats during active imaging passes.	2	4	8	Dedicated copper thermal strap to outer radiating structural panel.	1	4	4	Thermal Vacuum
RS-20	EPS	Solar array deployment shock damages solar cell interconnections.	3	3	9	Deploy with active dashpots; damp deployment with rubber stops.	1	3	3	Shock Test

d

References

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- [2] **ECSS-E-ST-33-11C**, *Space Engineering: Electrical and Electronic*, European Cooperation for Space Standardization, 2021.
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Appendix

موضوعات زیر بهتر است به جای نوشته شدن در متن اصلی گزارش، در بخش پیوست‌ها آورده شوند:

- 1- اثبات‌های ریاضی یا عملیات ریاضی طولانی.
 - 2- داده و اطلاعات نمونه(های) مورد مطالعه (Case Study) چنانچه طولانی باشد.
 - 3- نتایج کارهای دیگران چنانچه نیاز به تفصیل باشد.
 - 4- مجموعه تعاریف متغیرها و پارامترها، چنانچه طولانی بوده و در متن به انجام نرسیده باشد.
- برای شماره‌گذاری روابط، جداول و اشکال موجود در پیوست از ساختار متفاوتی نسبت به متن اصلی استفاده می‌شود که در زیر به‌عنوان نمونه نمایش داده شده‌است.

$$f = m a \quad (\text{پ-1})$$

جدول پ-1: متن کد بدنه اصلی نرم‌افزار تحلیل آیرودینامیکی.

```
01 program AeroPack;
02 uses
03   Forms,
04   Unit1 in 'Unit1.pas' {Form1},
05   Dialogs,
06   Sysutils;
07 {$R *.res}
08 begin
09   Application.Initialize;
10   Application.Title := 'AeroPack';
11   Application.CreateForm(TForm1, Form1);
12   if pos('/h',Form1.Switches)<>0 then
13     begin
14       Application.ShowMainForm:=False;
15       Form1.Visible:=False;
16     end;
17   Application.Run;
18 end.
```

در صورتیکه سوئیچ /h در رشته سوئیچ موجود باشد، متغیر ShowMainForm و خصوصیت Visible فرم اصلی را برابر با False قرار می‌دهد. نتیجه این کار عدم نمایش فرم اصلی خواهد بود.

شکل پ-1: بدنه اصلی نرم‌افزار تحلیل آیرودینامیکی.

